

REAL ESTATE PRICE PREDICTION IN MACHINE LEARNING

A Project Report Submitted to the Madurai Kamaraj University

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BACHELOR OF SCIENCE IN COMPUTER SCIENCE

Submitted by

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BONAFIDE CERTIFICATE

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PROJECT GUIDE

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INTERNAL EXAMINER

EXTERNAL EXMINER

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DECLARATION

I hereby declare that this work was carried out by me under the guidance of **Mrs.R.SHIYAMALA M.Sc., M.Phil.,** Assistant Professor in the Department of Bachelor of Science in Computer science. The period of project work is from June 2025 To October 2025. This project work is submitted to **P.K.N. Arts & Science College, Department of Bachelor of Science in Computer Science** in partial fulfillment for the requirement.

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REAL ESTATE PRICE PREDICTION IN MACHINE LEARNING

S.NO	CONTENTS	PAGE NO
1	Synopsis	1`
2	Introduction	2
	2.1.Overview of the project	
3	System Analysis	4
	3.1.Existing System	
	3.2.Proposed System	5
	3.3.Feasibility Study	6
4	System Specification	7
	4.1.Hardware Specification	
	4.2.Software Specification	8
	4.3. Software Description	
5	System Design	21
	5.1 Structure Charts	
	5.2 Table Structure	23
6	System Testing	27
	6.1.Unit Testing	
	6.2.White box Testing	
	6.3.Black box Testing	
7	Module Specification	29
8	System Implementation	32

9	Conclusion	36
10	Future Enhancement	37
11	Appendix	38
	11.1.Sample Coding	
	11.2.Screen Shots	44
12	References & Bibliography	50

1 ABSTRACT

The real estate market is one of the most competitive and price-focused in existence today. With their budgets and an analysis of market tactics, people are searching to purchase a home. Creating a precise model for projecting house prices is always necessary for socioeconomic development and the welfare of citizens. such that a real estate agent or a home seller or buyer can get a gut feeling for using the model to make informed selections. The primary drawback of the existing method is that it determines the price of a house without the essential foresight into potential future market movements, which leads to an increase in price. So, the primary goal of our study is to apply machine learning to accurately anticipate the price of a house. For the purpose of estimating home prices and attempting to provide clients with effective house pricing that respects both their goals and their budget, numerous elements must be taken into account. In the proposed methodology, it is shown how to apply a machine learning algorithm to predict real estate/house prices using Kaggle dataset. This study addresses the application of machine learning algorithm for predicting real estate prices. To be more precise, a Random Forest Regressor model is used to predict a property's price depending on several characteristics like location, size, number of rooms, etc.. A publicly accessible dataset with details on real estate properties in a certain area was utilised to train and test the model. The findings demonstrate that the Random Forest Regressor model can reasonably predict real estate prices. According to the study's findings, machine learning algorithms can be used to accurately anticipate real estate prices, offering beneficial information to buyers, sellers, and real estate professionals.

2. INTRODUCTION

- The project "Real Estate Price Prediction using Random Forest Regressor" aims to develop a machine learning model that can accurately predict the prices of real estate properties based on a variety of relevant features, such as location, square footage, and number of bedrooms. The real estate industry is one of the largest and most important industries in the world, and accurate price prediction can be critical for buying, selling, and investing in real estate.
- To achieve this goal, the project utilizes a Random Forest Regressor model, which is a powerful machine learning algorithm that has been shown to perform well in many prediction tasks. The model is trained on a dataset of real estate properties, which includes a variety of features such as the number of bedrooms and bathrooms, square footage, and location.
- By accurately predicting real estate prices, this project has the potential to make buying, selling, and investing in real estate more efficient and profitable. Additionally, the project may have broader implications for the use of machine learning in the real estate industry, as it demonstrates the power of these techniques for solving complex prediction tasks.

2.1 OVERVIEW

THE GENERAL WORKFLOW INVOLVES THE FOLLOWING STEP:

- Data Collection: Gather property data including relevant features like number of bedrooms, bathrooms, square footage, location, and amenities.?
- Data Preprocessing: Clean the data, handle missing values, normalize numerical features, and encode categorical features as needed.?
- Exploratory Data Analysis (EDA): Visualize data distributions, analyze relationships between features and price, and identify significant variables using libraries like Pandas, Matplotlib, and Seaborn.?
- Feature Selection: Identify which features are most important for predicting prices.

- **Model Building:** Choose regression algorithms—commonly linear regression, random forests, or gradient boosting techniques like XGBoost—and split the dataset into training and test sets.?
- **Model Training & Evaluation:** Train the model and evaluate using metrics like mean squared error (MSE), mean absolute error (MAE), and R-squared to assess accuracy.?
- **Prediction & Deployment:** Use the trained model to predict property prices and, if needed, integrate it into real-world systems for automated evaluation.
- **Pandas and NumPy:** For data manipulation and numerical operations.?
- **Matplotlib and Seaborn:** For data visualization and exploratory analysis.?
- **Scikit-learn:** For splitting data, building, and evaluating machine learning models.?
- **XGBoost:** For high-performance gradient boosting regression.

3. SYSTEM ANALYSIS

3.1 EXISTING SYSTEM:

- Ahmed et al. design a neural network-based model for predicting housing Market Performance. This model is trained through a historical market performance dataset to predict unforeseen future performances. The model testing and validation show that the error in predicting his Neural Net is in the range between -2 and $+2$ percent.

- To predict the Singapore housing market, Lim et al. design neural networks. They used two algorithms for prediction, the multilayer perceptron, and autoregressive integrated moving average. The model with high accuracy score is used for prediction, and the model with lower mean square error (MSE) of the ANN models shows that ANN is best over other predictive tools.

- Chica et al. designed Cokriging a Multivariate Spatial Method for predicting Housing Location Price. This method estimates correlated spatial variables, interpolated maps of house prices are created, providing information about house location prices to appraisers and real estate agents. During the experiment, housing location price prediction value is estimated using methods: isotopic data cokriging and heterotopic data cokriging. Results from both methods are compared, and prediction from the best method is selected.

DISADVANTAGES OF EXISTING SYSTEM:

- Complexity: Neural network-based models can be very complex and difficult to interpret, especially when dealing with large amounts of data. This can make it challenging to understand how the model is making its predictions, which could be a disadvantage when working with real estate data.

- Overfitting: Neural networks are prone to overfitting, which occurs when the model becomes too closely fitted to the training data and performs poorly on new data. This can be a particular challenge when working with real estate data, which can be highly variable and unpredictable.

- Data requirements: Neural networks require a large amount of data to be effective. When working with real estate data, it may be difficult to obtain a sufficiently large dataset with enough features to make accurate predictions.

- Computational resources: Neural networks require significant computational resources to train

and run, which can be a disadvantage for smaller organizations or those with limited resources.

- **Difficulty of implementation:** Implementing a neural network-based model can be complex, and may require significant technical expertise. This can be a disadvantage for organizations or individuals who do not have access to this expertise.

3.2 PROPOSED SYSTEM:

- In the proposed system, we concentrate on using the machine learning technique Random Forest Regressor model to estimate real estate/house price. We developed the "Real Estate Price Prediction using Machine Learning" algorithm and used several features to forecast home prices. With the help of the previously collected data, we are able to build a model for the proposed system. Of this data, 80% is utilised for training and the remaining 20% is used for testing. Here, a ".csv" file is used to hold the raw data. To overcome these difficulties, we primarily used two machine learning packages. Pandas was the first, while Scikit-Learn is the second. The ".csv" file was loaded into the Jupiter notebook using pandas, which was also used to clean and alter the data. Another was sklearn, which was used for actual analysis and offers a number of built-in functions that aid with issue solving.

- With the help of the numerous features that the user inputs, the suggested system aims to estimate the cost of a house. The ML model receives these features, and depending on how they influence the label, it makes a prediction. This will be accomplished by initially looking for a dataset that is suitable for both the developer's and the user's demands. Also, the dataset will undergo a procedure known as data cleaning after it has been finalised, during which all unnecessary data will be removed and the raw data will be converted into a.csv file. Also, the data will undergo data preprocessing, during which any missing data will be resolved and, if necessary, label encoding will be performed. Also, this will go through data transformation and be turned into a NumPy array before being sent to train the model. An algorithm and model that can make reliable predictions will be concluded by extracting the error rate of the Random Forest Regressor model used to train the model.

- The user can log in and then fill out a form with information about different aspects of their property that they wish to anticipate the price of. Also, the form will be submitted after carefully choosing the qualities. The user's data will then be sent to the model, where they can view the estimated price of the property they input in a matter of seconds

ADVANTAGES OF PROPOSED SYSTEM:

- **Robustness:** Random Forest models are highly robust to outliers, missing data, and other types of noise in the dataset. This makes them well-suited for real estate price prediction, where there can be significant variability in the data.
- **Interpretability:** Unlike some other machine learning models, Random Forests are relatively easy to interpret. The feature importances can be used to identify the most important factors that are driving the prediction, which can be useful for decision-making.
- **Accuracy:** Random Forest models have been shown to be highly accurate in many prediction tasks, and have been found to perform well in real estate price prediction specifically.
- **Scalability:** Random Forest models are highly scalable, and can handle large datasets with many features. This makes them well-suited for real estate price prediction, which often involves working with large amounts of data.
- **Flexibility:** Random Forest models can be used for both regression and classification tasks, making them a versatile tool for real estate price prediction and related tasks. Additionally, they can be combined with other models, such as gradient boosting, to improve performance even further.

3.3 FEASIBILITY STUDY:

- **Data Availability:** Sufficient, high-quality real estate data (with features like location, size, amenities, and market trends) is crucial. Public datasets from platforms like Kaggle, governmental sources, and commercial APIs are often used.
- **Technical Infrastructure:** Python's ecosystem thoroughly supports data analysis, model building, and deployment, making technical implementation feasible for researchers and businesses.
- **Model Accuracy:** Advanced regression models—linear regression, Random Forest, XGBoost—typically achieve high accuracy if the dataset is cleaned and features are correctly selected and engineered.
- **Cost & Efficiency:** The development cost is relatively low since Python and associated libraries are open-source. The modeling process can be automated for scalable deployment in web or mobile applications.

Implementation Challenges:

- Real estate price prediction can be complicated by missing data, feature correlation, and external economic factors.
- Location-specific models may require regional data, making generic solutions less accurate for highly localized markets.

4. SYSTEM SPECIFICATION

4.1 HARDWARE SPECIFICATION:

- System : Pentium i3 Processor.
- Hard Disk : 500 GB.
- Monitor : 15'' LED
- Input Devices : Keyboard, Mouse
- Ram : 4 GB

4.2 SOFTWARE SPECIFICATION:

- Operating system : Windows 10.
- Coding Language : Python.
- Web Framework : Flask.

4.3 SOFTWARE DESCRIPTION:

What is Machine Learning?

- Machine Learning is a system of computer algorithms that can learn from example through self-improvement without being explicitly coded by a programmer. Machine learning is a part of artificial Intelligence which combines data with statistical tools to predict an output which can be used to make actionable insights.
- The breakthrough comes with the idea that a machine can singularly learn from the data (i.e., example) to produce accurate results. Machine learning is closely related to data mining and Bayesian predictive modeling. The machine receives data as input and uses an algorithm to formulate answers.
- A typical machine learning tasks are to provide a recommendation. For those who have a Netflix account, all recommendations of movies or series are based on the user's historical

data. Tech companies are using unsupervised learning to improve the user experience with personalizing recommendation.

- Machine learning is also used for a variety of tasks like fraud detection, predictive maintenance, portfolio optimization, automatize task and so on.

Machine Learning vs. Traditional Programming

- Traditional programming differs significantly from machine learning. In traditional programming, a programmer code all the rules in consultation with an expert in the industry for which software is being developed. Each rule is based on a logical foundation; the machine will execute an output following the logical statement. When the system grows complex, more rules need to be written. It can quickly become unsustainable to maintain.
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Traditional Programming

Machine learning is supposed to overcome this issue. The machine learns how the input and output data are correlated and it writes a rule. The programmers do not need to write new rules each time there is new data. The algorithms adapt in response to new data and experiences to improve efficacy over time.

MACHINE LEARNING

How does Machine Learning Work?

- Machine learning is the brain where all the learning takes place. The way the machine learns is similar to the human being. Humans learn from experience. The more we know, the more easily we can predict. By analogy, when we face an unknown situation, the likelihood of success is lower than the known situation. Machines are trained the same. To make an

accurate prediction, the machine sees an example. When we give the machine a similar example, it can figure out the outcome. However, like a human, if its feed a previously unseen example, the machine has difficulties to predict.

- The core objective of machine learning is the learning and inference. First of all, the machine learns through the discovery of patterns. This discovery is made thanks to the data. One crucial part of the data scientist is to choose carefully which data to provide to the machine. The list of attributes used to solve a problem is called a feature vector. You can think of a feature vector as a subset of data that is used to tackle a problem.
- The machine uses some fancy algorithms to simplify the reality and transform this discovery into a model. Therefore, the learning stage is used to describe the data and summarize it into a model.
- For instance, the machine is trying to understand the relationship between the wage of an individual and the likelihood to go to a fancy restaurant. It turns out the machine finds a positive relationship between wage and going to a high-end restaurant:
- When the model is built, it is possible to test how powerful it is on never-seen-before data. The new data are transformed into a features vector, go through the model and give a prediction. This is all the beautiful part of machine learning. There is no need to update the rules or train again the model. You can use the model previously trained to make inference on new data.

The life of Machine Learning programs is straightforward and can be summarized in the following points:

1. Define a question
2. Collect data
3. Visualize data
4. Train algorithm
5. Test the Algorithm
6. Collect feedback

7. Refine the algorithm
8. Loop 4-7 until the results are satisfying
9. Use the model to make a prediction

Once the algorithm gets good at drawing the right conclusions, it applies that knowledge to new sets of data.

Machine Learning Algorithms and Where they are Used?

Machine learning Algorithms

Machine learning can be grouped into two broad learning tasks: Supervised and Unsupervised.

There are many other algorithms

Supervised learning

An algorithm uses training data and feedback from humans to learn the relationship of given inputs to a given output. For instance, a practitioner can use marketing expense and weather forecast as input data to predict the sales of cans.

You can use supervised learning when the output data is known. The algorithm will predict new data.

There are two categories of supervised learning:

- Classification task
- Regression task

Classification

- Imagine you want to predict the gender of a customer for a commercial. You will start gathering data on the height, weight, job, salary, purchasing basket, etc. from your customer database. You know the gender of each of your customer, it can only be male or female. The objective of the classifier will be to assign a probability of being a male or a female (i.e., the label) based on the information (i.e., features you have collected). When the model learned how to recognize male or female, you can use new data to make a prediction. For instance, you just got new information from an unknown customer, and you want to know if it is a male or female. If the classifier predicts male = 70%, it means the algorithm is sure at 70% that this customer is a male, and 30% it is a female.
- The label can be of two or more classes. The above Machine learning example has only two classes, but if a classifier needs to predict object, it has dozens of classes

(e.g.glass,table,shoes, etc. each object represents a class)

Regression

When the output is a continuous value, the task is a regression. For instance, a financial analyst may need to forecast the value of a stock based on a range of feature like equity, previous stock performances, macroeconomics index. The system will be trained to estimate the price of the stocks with the lowest possible error.

- ❖ Algorithm Name
- ❖ Description
- ❖ Type
- ❖ Linear regression
- ❖ Finds a way to correlate each feature to the output to help predict future values.
- ❖ Regression
- ❖ Logistic regression
- ❖ Extension of linear regression that's used for classification tasks. The output variable is binary (e.g., only black or white) rather than continuous (e.g., an infinite list of potential colors)
- ❖ Classification
- ❖ Decision tree
- ❖ Highly interpretable classification or regression model that splits data-feature values into branches at decision nodes (e.g., if a feature is a color, each possible color becomes a new branch) until a final decision output is made
- ❖ Regression Classification
- ❖ Naive Bayes
- ❖ The Bayesian method is a classification method that makes use of the Bayesian theorem. The theorem updates the prior knowledge of an event with the independent probability of each feature that can affect the event.
- ❖ Regression Classification
- ❖ Support vector machine
- ❖ Support Vector Machine, or SVM, is typically used for the classification task. SVM algorithm finds a hyperplane that optimally divided the classes. It is best used with a non-linear solver.
- ❖ Regression (not very common) Classification

Machine Learning (ML) algorithm:

- There are plenty of machine learning algorithms. The choice of the algorithm is based on the objective.
- In the Machine learning example below, the task is to predict the type of flower among the three varieties. The predictions are based on the length and the width of the petal. The picture depicts the results of ten different algorithms. The picture on the top left is the dataset. The data is classified into three categories: red, light blue and dark blue. There are some groupings. For instance, from the second image, everything in the upper left belongs to the red category, in the middle part, there is a mixture of uncertainty and light blue while the bottom corresponds to the dark category. The other images show different algorithms and how they try to classified the data.

Challenges and Limitations of Machine Learning

The primary challenge of machine learning is the lack of data or the diversity in the dataset. A machine cannot learn if there is no data available. Besides, a dataset with a lack of diversity gives the machine a hard time. A machine needs to have heterogeneity to learn meaningful insight. It is rare that an algorithm can extract information when there are no or few variations. It is recommended to have at least 20 observations per group to help the machine learn. This constraint leads to poor evaluation and prediction.

Application of Machine Learning

Augmentation:

- Machine learning, which assists humans with their day-to-day tasks, personally or commercially without having complete control of the output. Such machine learning is used in different ways such as Virtual Assistant, Data analysis, software solutions. The primary user is to reduce errors due to human bias.

Automation:

- Machine learning, which works entirely autonomously in any field without the need for any human intervention. For example, robots performing the essential process steps in manufacturing plants.

Finance Industry

- Machine learning is growing in popularity in the finance industry. Banks are mainly using ML to find patterns inside the data but also to prevent fraud.

Government organization

- The government makes use of ML to manage public safety and utilities. Take the example of China with the massive face recognition. The government uses Artificial intelligence to prevent jaywalker.

Healthcare industry

- Healthcare was one of the first industry to use machine learning with image detection.

Marketing

- Broad use of AI is done in marketing thanks to abundant access to data. Before the age of mass data, researchers develop advanced mathematical tools like Bayesian analysis to estimate the value of a customer. With the boom of data, marketing department relies on AI to optimize the customer relationship and marketing campaign.

Example of application of Machine Learning in Supply Chain

- Machine learning gives terrific results for visual pattern recognition, opening up many potential applications in physical inspection and maintenance across the entire supply chain network.
- Unsupervised learning can quickly search for comparable patterns in the diverse dataset. In turn, the machine can perform quality inspection throughout the logistics hub, shipment with damage and wear.
- For instance, IBM's Watson platform can determine shipping container damage. Watson combines visual and systems-based data to track, report and make recommendations in real-time.
- In past year stock manager relies extensively on the primary method to evaluate and forecast the inventory. When combining big data and machine learning, better forecasting techniques have been implemented (an improvement of 20 to 30 % over traditional forecasting tools). In term of sales, it means an increase of 2 to 3 % due to the potential reduction in inventory costs.

Example of Machine Learning Google Car

- For example, everybody knows the Google car. The car is full of lasers on the roof which are telling it where it is regarding the surrounding area. It has radar in the front, which is informing the car of the speed and motion of all the cars around it. It uses all of that data to figure out not only how to drive the car but also to figure out and predict what potential

drivers around the car are going to do. What's impressive is that the car is processing almost a gigabyte a second of data.

Why is Machine Learning Important?

- Machine learning is the best tool so far to analyze, understand and identify a pattern in the data. One of the main ideas behind machine learning is that the computer can be trained to automate tasks that would be exhaustive or impossible for a human being. The clear breach from the traditional analysis is that machine learning can take decisions with minimal human intervention.
- Take the following example for this ML tutorial; a retail agent can estimate the price of a house based on his own experience and his knowledge of the market.
- A machine can be trained to translate the knowledge of an expert into features. The features are all the characteristics of a house, neighborhood, economic environment, etc. that make the price difference. For the expert, it took him probably some years to master the art of estimate the price of a house. His expertise is getting better and better after each sale.
- For the machine, it takes millions of data, (i.e., example) to master this art. At the very beginning of its learning, the machine makes a mistake, somehow like the junior salesman. Once the machine sees all the example, it got enough knowledge to make its estimation. At the same time, with incredible accuracy. The machine is also able to adjust its mistake accordingly.
- Most of the big company have understood the value of machine learning and holding data. McKinsey have estimated that the value of analytics ranges from \$9.5 trillion to \$15.4 trillion while \$5 to 7 trillion can be attributed to the most advanced AI techniques.
- Machine learning (ML) is the study of computer algorithms that improve automatically through experience. It is seen as a part of artificial intelligence. Machine learning algorithms build a model based on sample data, known as "training data", in order to make predictions or decisions without being explicitly programmed to do so. Machine learning algorithms are used in a wide variety of applications, such as email filtering and computer vision, where it is difficult or unfeasible to develop conventional algorithms to perform the needed tasks.
- A subset of machine learning is closely related to computational statistics, which focuses on making predictions using computers; but not all machine learning is statistical learning. The study of mathematical optimization delivers methods, theory and application domains to the field of machine learning. Data mining is a related field of study, focusing on exploratory

data analysis through unsupervised learning. In its application across business problems, machine learning is also referred to as predictive analytics.

- Machine learning approaches are traditionally divided into three broad categories, depending on the nature of the "signal" or "feedback" available to the learning system:

Supervised learning:

- The computer is presented with example inputs and their desired outputs, given by a "teacher", and the goal is to learn a general rule that maps inputs to outputs.
- Unsupervised learning: No labels are given to the learning algorithm, leaving it on its own to find structure in its input. Unsupervised learning can be a goal in itself (discovering hidden patterns in data) or a means towards an end (feature learning)..

History and relationships to other fields

- The term machine learning was coined in 1959 by Arthur Samuel, an American IBMer and pioneer in the field of computer gaming and artificial intelligence. A representative book of the machine learning research during the 1960s was the Nilsson's book on Learning Machines, dealing mostly with machine learning for pattern classification. Interest related to pattern recognition continued into the 1970s, as described by Duda and Hart in 1973. In 1981 a report was given on using teaching strategies so that a neural network learns to recognize 40 characters (26 letters, 10 digits, and 4 special symbols) from a computer terminal.
- Tom M. Mitchell provided a widely quoted, more formal definition of the algorithms studied in the machine learning field: "A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T, as measured by P, improves with experience E." This definition of the tasks in which machine learning is concerned offers a fundamentally operational definition rather than defining the field in cognitive terms. This follows Alan Turing's proposal in his paper "Computing Machinery and Intelligence", in which the question "Can machines think?" is replaced with the question "Can machines do what we (as thinking entities) can do?".
- Modern day machine learning has two objectives, one is to classify data based on models which have been developed, the other purpose is to make predictions for future outcomes based on these models. A hypothetical algorithm specific to classifying data may use

computer vision of moles coupled with supervised learning in order to train it to classify the cancerous moles. Where as, a machine learning algorithm for stock trading may inform the trader of future potential predictions.

ARTIFICIAL INTELLIGENCE

Machine Learning as subfield of AI

- Part of Machine Learning as subfield of AI or part of AI as subfield of Machine Learn. As a scientific endeavor, machine learning grew out of the quest for artificial intelligence. In the early days of AI as an academic discipline, some researchers were interested in having machines learn from data. They attempted to approach the problem with various symbolic methods, as well as what was then termed "neural networks"; these were mostly perceptrons and other models that were later found to be reinventions of the generalized linear models of statistics. Probabilistic reasoning was also employed, especially in automated medical diagnosis.
- However, an increasing emphasis on the logical, knowledge-based approach caused a rift between AI and machine learning. Probabilistic systems were plagued by theoretical and practical problems of data acquisition and representation. By 1980, expert systems had come to dominate AI, and statistics was out of favor. Work on symbolic/knowledge-based learning did continue within AI, leading to inductive logic programming, but the more statistical line of research was now outside the field of AI proper, in pattern recognition and information retrieval. Neural networks research had been abandoned by AI and computer science around the same time. This line, too, was continued outside the AI/CS field, as "connectionism", by researchers from other disciplines including Hopfield, Rumelhart and Hinton. Their main success came in the mid-1980s with the reinvention of backpropagation.
- Machine learning (ML), reorganized as a separate field, started to flourish in the 1990s. The field changed its goal from achieving artificial intelligence to tackling solvable problems of a practical nature. It shifted focus away from the symbolic approaches it had inherited from AI, and toward methods and models borrowed from statistics and probability theory.
- As of 2020, many sources continue to assert that machine learning remains a subfield of AI. The main disagreement is whether all of ML is part of AI, as this would mean that anyone using ML could claim they are using AI. Others have the view that not all of ML is part of AI where only an 'intelligent' subset of ML is part of AI.

- The question to what is the difference between ML and AI is answered by Judea Pearl in The Book of Why. Accordingly ML learns and predicts based on passive observations, whereas AI implies an agent interacting with the environment to learn and take actions that maximize its chance of successfully achieving its goals.

Data mining

- Machine learning and data mining often employ the same methods and overlap significantly, but while machine learning focuses on prediction, based on known properties learned from the training data, data mining focuses on the discovery of (previously) unknown properties in the data (this is the analysis step of knowledge discovery in databases). Data mining uses many machine learning methods, but with different goals; on the other hand, machine learning also employs data mining methods as "unsupervised learning" or as a preprocessing step to improve learner accuracy. Much of the confusion between these two research communities (which do often have separate conferences and separate journals, ECML PKDD being a major exception) comes from the basic assumptions they work with: in machine learning, performance is usually evaluated with respect to the ability to reproduce known knowledge, while in knowledge discovery and data mining (KDD) the key task is the discovery of previously unknown knowledge. Evaluated with respect to known knowledge, an uninformed (unsupervised) method will easily be outperformed by other supervised methods, while in a typical KDD task, supervised methods cannot be used due to the unavailability of training data.

Optimization

Machine learning also has intimate ties to optimization: many learning problems are formulated as minimization of some loss function on a training set of examples. Loss functions express the discrepancy between the predictions of the model being trained and the actual problem instances (for example, in classification, one wants to assign a label to instances, and models are trained to correctly predict the pre-assigned labels of a set of examples).

Generalization

The difference between optimization and machine learning arises from the goal of generalization: while optimization algorithms can minimize the loss on a training set, machine

learning is concerned with minimizing the loss on unseen samples. Characterizing the generalization of various learning algorithms is an active topic of current research, especially for deep learning algorithms.

Supervised learning

- A support vector machine is a supervised learning model that divides the data into regions separated by a linear boundary. Here, the linear boundary divides the black circles from the white.
- Supervised learning algorithms build a mathematical model of a set of data that contains both the inputs and the desired outputs. The data is known as training data, and consists of a set of training examples. Each training example has one or more inputs and the desired output, also known as a supervisory signal. In the mathematical model, each training example is represented by an array or vector, sometimes called a feature vector, and the training data is represented by a matrix. Through iterative optimization of an objective function, supervised learning algorithms learn a function that can be used to predict the output associated with new inputs. An optimal function will allow the algorithm to correctly determine the output for inputs that were not a part of the training data. An algorithm that improves the accuracy of its outputs or predictions over time is said to have learned to perform that task.
- Types of supervised learning algorithms include active learning, classification and regression. Classification algorithms are used when the outputs are restricted to a limited set of values, and regression algorithms are used when the outputs may have any numerical value within a range. As an example, for a classification algorithm that filters emails, the input would be an incoming email, and the output would be the name of the folder in which to file the email.
- Similarity learning is an area of supervised machine learning closely related to regression and classification, but the goal is to learn from examples using a similarity function that measures how similar or related two objects are. It has applications in ranking, recommendation systems, visual identity tracking, face verification, and speaker verification.

Artificial neural networks

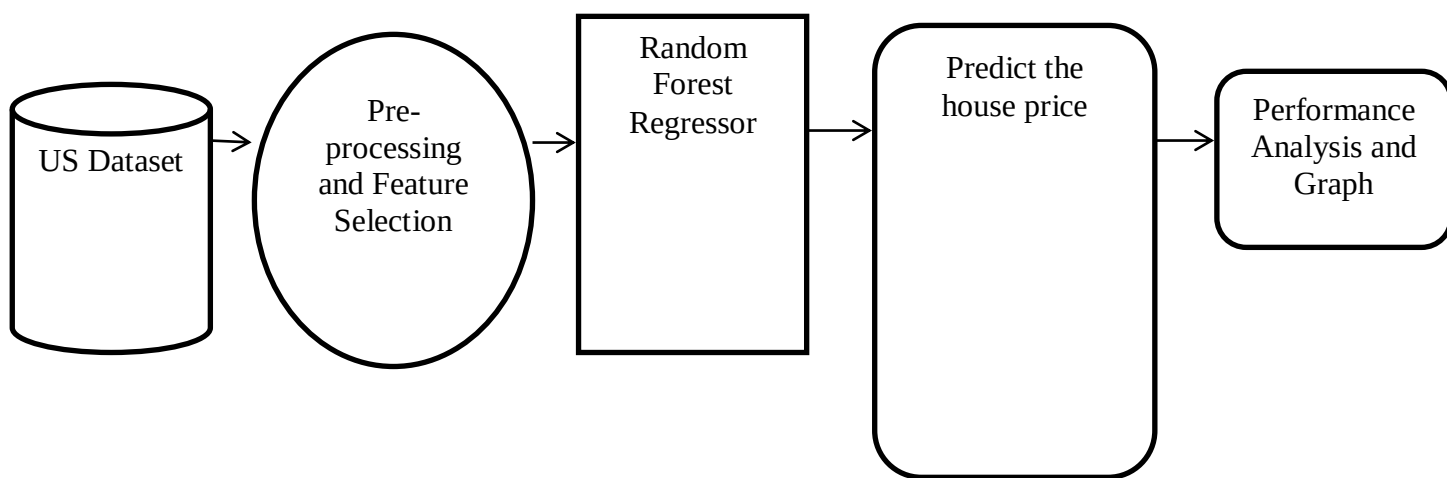
- An artificial neural network is an interconnected group of nodes, akin to the vast network of neurons in a brain. Here, each circular node represents an artificial neuron and an arrow represents a connection from the output of one artificial neuron to the input of another.
- Artificial neural networks (ANNs), or connectionist systems, are computing systems vaguely inspired by the biological neural networks that constitute animal brains. Such systems "learn" to perform tasks by considering examples, generally without being programmed with any task-specific rules.
- An ANN is a model based on a collection of connected units or nodes called "artificial neurons", which loosely model the neurons in a biological brain. Each connection, like the synapses in a biological brain, can transmit information, a "signal", from one artificial neuron to another. An artificial neuron that receives a signal can process it and then signal additional artificial neurons connected to it. In common ANN implementations, the signal at a connection between artificial neurons is a real number, and the output of each artificial neuron is computed by some non-linear function of the sum of its inputs. The connections between artificial neurons are called "edges". Artificial neurons and edges typically have a weight that adjusts as learning proceeds. The weight increases or decreases the strength of the signal at a connection. Artificial neurons may have a threshold such that the signal is only sent if the aggregate signal crosses that threshold. Typically, artificial neurons are aggregated into layers. Different layers may perform different kinds of transformations on their inputs. Signals travel from the first layer (the input layer) to the last layer (the output layer), possibly after traversing the layers multiple times.
- The original goal of the ANN approach was to solve problems in the same way that a human brain would. However, over time, attention moved to performing specific tasks, leading to deviations from biology. Artificial neural networks have been used on a variety of tasks, including computer vision, speech recognition, machine translation, social network filtering, playing board and video games and medical diagnosis.
- Deep learning consists of multiple hidden layers in an artificial neural network. This approach tries to model the way the human brain processes light and sound into vision and hearing. Some successful applications of deep learning are computer vision and speech recognition.

Regression analysis

Regression analysis encompasses a large variety of statistical methods to estimate the relationship between input variables and their associated features. Its most common form is linear regression, where a single line is drawn to best fit the given data according to a mathematical criterion such as ordinary least squares. The latter is often extended by regularization (mathematics) methods to mitigate overfitting and bias, as in ridge regression. When dealing with non-linear problems, go-to models include polynomial regression (for example, used for trendline fitting in Microsoft Excel, logistic regression (often used in statistical classification) or even kernel regression, which introduces non-linearity by taking advantage of the kernel trick to implicitly map input variables to higher-dimensional space.

5. SYSTEM DESIGN

5.1 SYSTEM ARCHITECTURE:



UML DIAGRAMS

- UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.
- The goal is for UML to become a common language for creating models of object oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.
- The Unified Modeling Language is a standard language for specifying, Visualization,

Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems.

- The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.
- The UML is a very important part of developing objects oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects

GOALS:

The Primary goals in the design of the UML are as follows:

- Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
- Provide extendibility and specialization mechanisms to extend the core concepts. Be independent of particular programming languages and development process.
- Provide a formal basis for understanding the modeling language.
- Encourage the growth of OO tools market.
- Support higher level development concepts such as collaborations, frameworks, patterns and components.
- Integrate best practices.

USE CASE DIAGRAM:

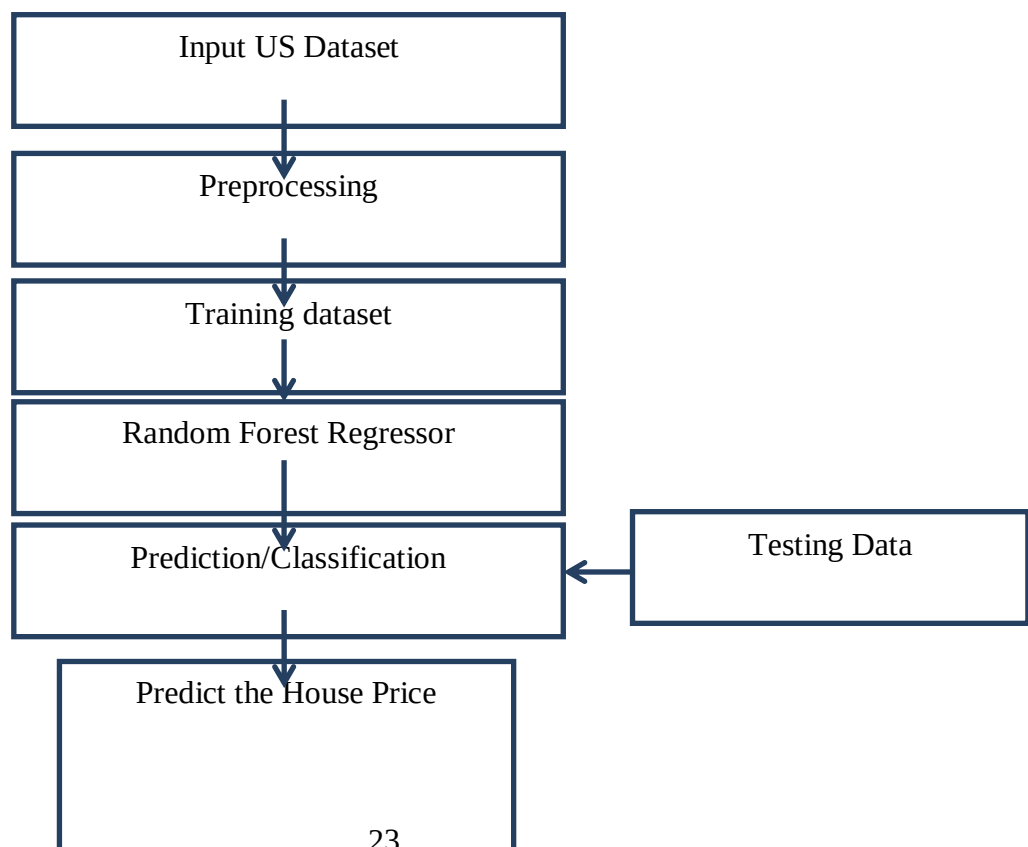
A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

5.2 TABLE STRUCTURE:

DATA FLOW DIAGRAM:

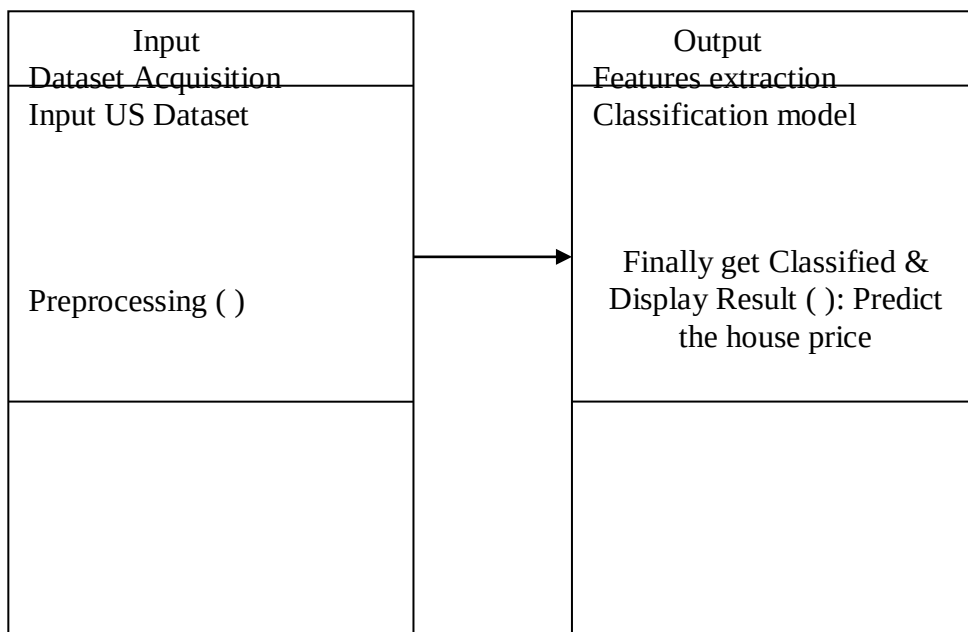
The DFD is also called as bubble chart. It is a simple graphical formalism that can be used to represent a system in terms of input data to the system, various processing carried out on this data, and the output data is generated by this system.

1. The data flow diagram (DFD) is one of the most important modeling tools. It is used to model the system components. These components are the system process, the data used by the process, an external entity that interacts with the system and the information flows in the system
2. DFD shows how the information moves through the system and how it is modified by a series of transformations. It is a graphical technique that depicts information flow and the transformations that are applied as data moves from input to output.
3. DFD is also known as bubble chart. A DFD may be used to represent a system at any level of abstraction. DFD may be partitioned into levels that represent increasing information flow and functional detail



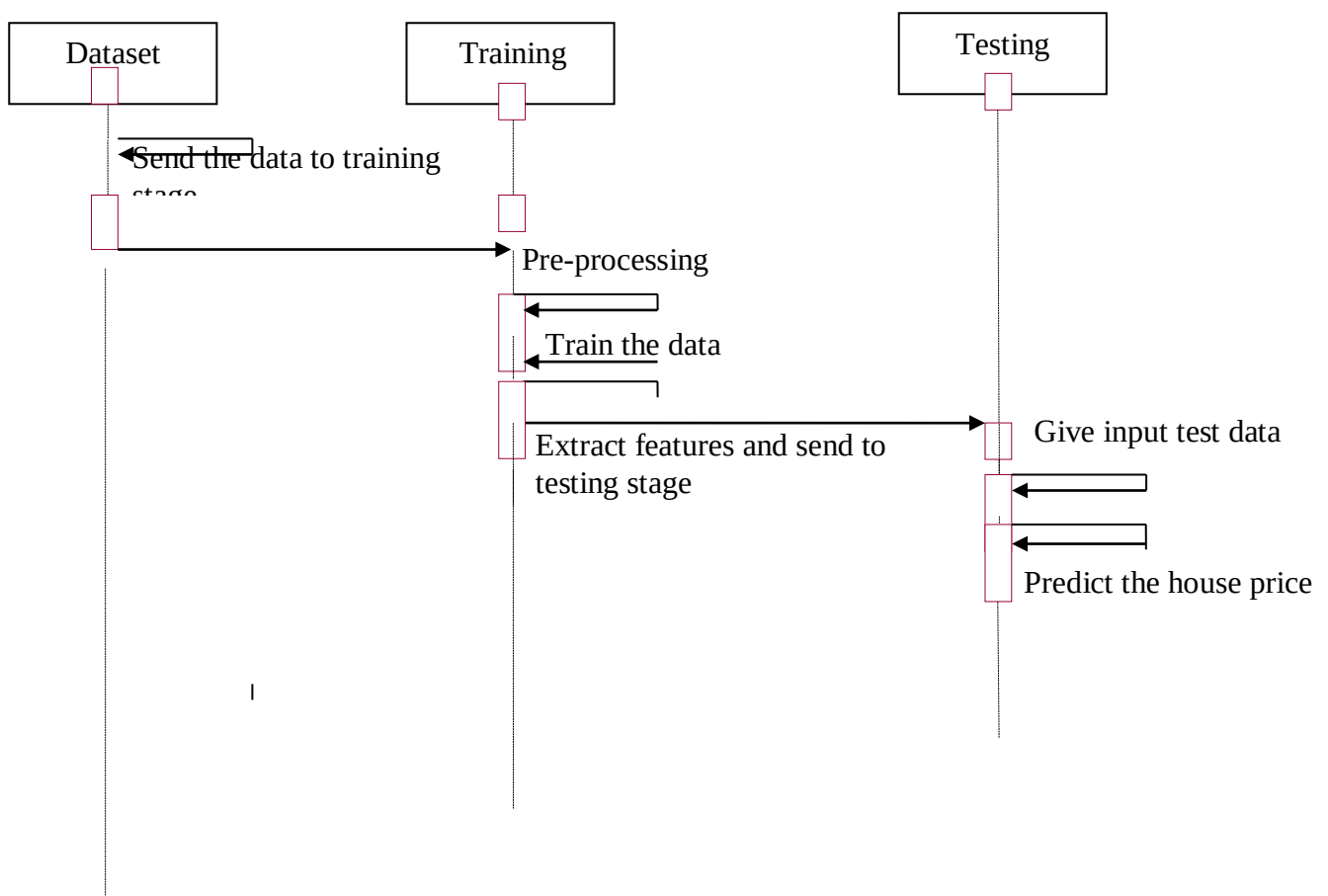
CLASS DIAGRAM:

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information.



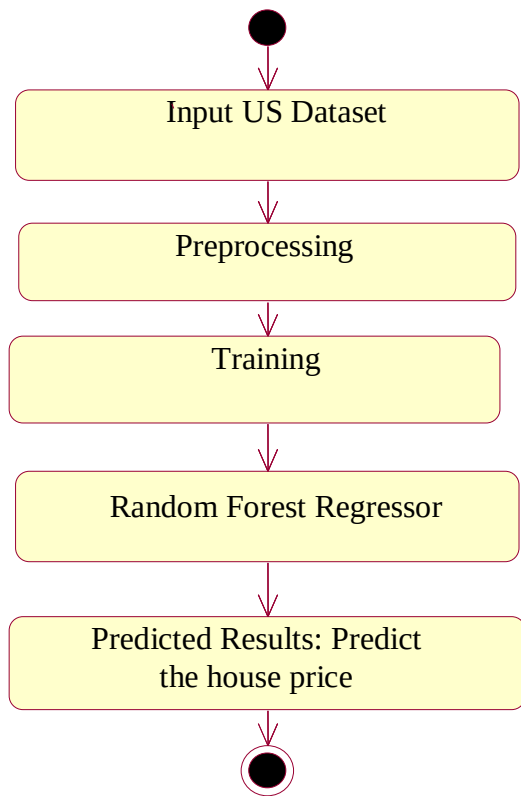
SEQUENCE DIAGRAM:

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.



ACTIVITY DIAGRAM:

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control.



6. SYSTEM TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub- assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

6.1 UNIT TESTING

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application. It is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

6.2 WHITE BOX TESTING

- White Box Testing is a testing in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is used to test areas that cannot be reached from a black box level.
- White box testing in this context focuses on verifying steps such as data preprocessing, feature engineering, model training, prediction logic, and code correctness. It includes unit testing for individual functions (like data cleaning or feature selection), integration tests for the model pipeline, and validation of output at each step.

6.3 BLACK BOX TESTING

- Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most

other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box. you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software work.

- Input real-world property data, such as location, square footage, number of bedrooms, and amenities into the model.
- Observe output prices and compare them with expected market values or benchmark datasets.
- Assess the quality of predictions using metrics like mean absolute error (MAE), root mean squared error (RMSE), or R-squared.
- Ensure predictions are consistent and realistic across a range of typical and edge-case input.

7. MODULE SPECIFICATION

MODULE 1: Data Collection Module

- **Objective:**
Collect and load real estate data into the system for analysis.
- **Tasks:**
 - Load dataset from CSV or Excel
 - Preview sample rows
 - Display dataset information (data types, null values, shape)
- **Technologies Used:**
Python, Pandas

MODULE 2: Data Preprocessing Module

- **Objective:**
Clean the raw data and prepare it for model training.
- **Tasks:**
 - Handle missing or inconsistent data
 - Encode categorical variables (e.g., location)
 - Normalize or scale numerical columns (optional)
 - Split into features (X) and target (y)
 - Train-test split
- **Technologies Used:**
Pandas, scikit-learn

MODULE 3: Model Training Module

- **Objective:**
Train a machine learning model to predict house prices.
- **Tasks:**
 - Train the model using regression algorithms
 - Tune hyperparameters (if needed)
 - Save the trained model for later use
- **Technologies Used:**
scikit-learn, joblib (or pickle)

MODULE 4: Model Evaluation Module

- **Objective:**
Evaluate how well the trained model performs.
- **Tasks:**
 - Generate predictions on test data
 - Calculate evaluation metrics (MAE, MSE, R^2 score)
 - Print or log results for analysis
- **Technologies Used:**
scikit-learn (metrics module), Python

MODULE 5: Prediction Module

- **Objective:**
Use the trained model to make price predictions for new input data.
- **Tasks:**
 - Load the trained model
 - Accept user input (area, bedrooms, location, etc.)
 - Make predictions and return results

- **Technologies Used:**
Python, joblib, NumPy

MODULE 6: Flask Web Interface Module

- **Objective:**
Create a web-based interface for users to interact with the model.
- **Tasks:**
 - Create input form (HTML) for property details
 - Display predicted price on result page
 - Handle routing using Flask (/, /predict)
- **Technologies Used:**
Flask, HTML/CSS (Jinja2 templates), Python

MODULE 7: Data Visualization Module (Optional but Recommended)

- **Objective:**
Visualize data and model results for better understanding.
- **Tasks:**
 - Plot price distribution
 - Show correlation matrix between features
 - Optional: Display graphs on web interface
- **Technologies Used:**
Matplotlib, Seaborn

8. SYSTEM IMPLEMENTATION

MODULES

- ❖ Data Collection
- ❖ Dataset
- ❖ Data Preparation
- ❖ Model Selection
- ❖ Analyze and Prediction
- ❖ Accuracy on test set
- ❖ Saving the Trained Model

MODULES DESCRIPTION:

Data Collection:

In the first module we develop the Data Collection part. This is the first real step towards the real development of a machine learning model, collecting data. This is a critical step that will cascade in how good the model will be, the more and better data that we get, the better our model will perform. There are several techniques to collect the data, like web scraping, manual interventions. Our dataset is located in the model folder. The datasets are referred from the popular dataset repository called kaggle. The following are the links of the dataset that we have referred.

Dataset Link:

<https://www.kaggle.com/datasets/jayaprakashpondy/us-real-estate-price>

Dataset:

The dataset consists of 4600 individual data. There are 18 columns in the dataset, which are described below.

- **'date'**- This probably refers to the date on which the information was collected.
- **'price'**- Price of the house
- **'bedrooms'**- No. of bedrooms in the house

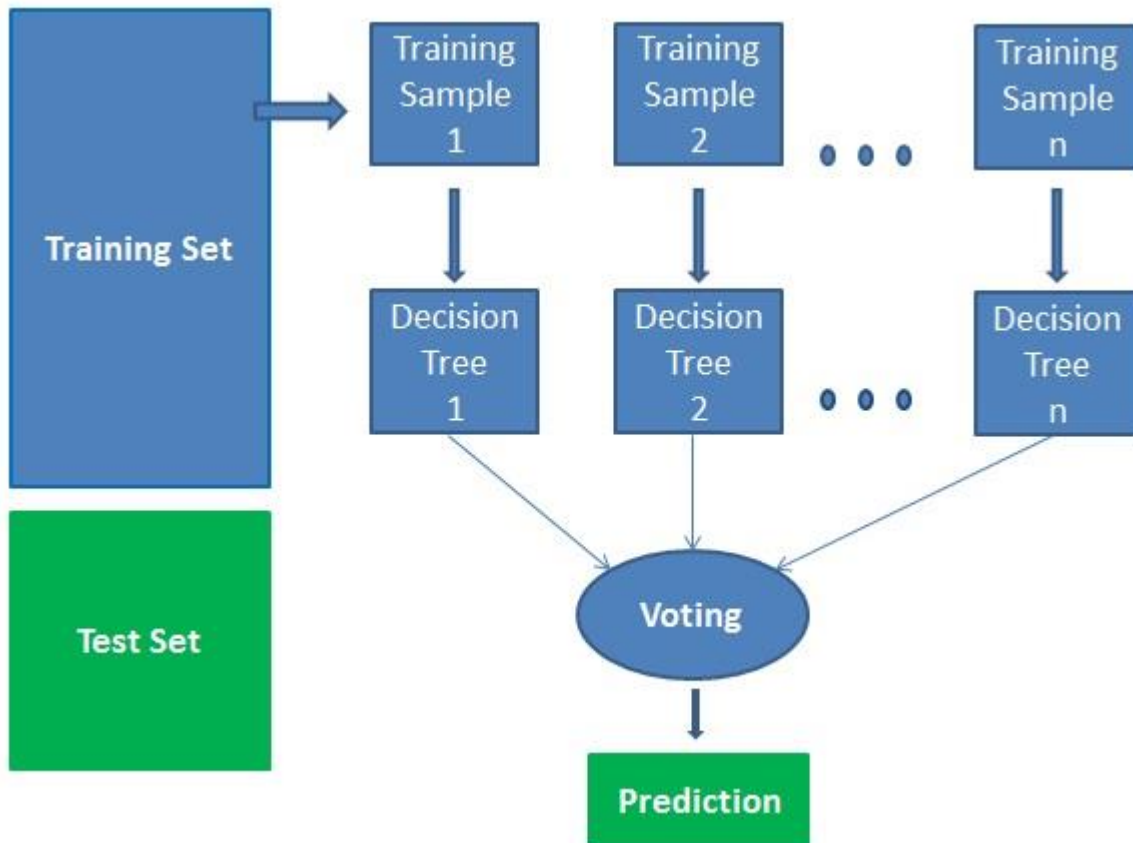
- **'bathrooms'**-No. of bathrooms in the house
- **'sqft_living'**- Area of the plot used for living
- **'sqft_lot'**- Total area of plot
- **'floors'**- No. of floors in the house
- **'waterfront'**- No. of waterfronts in the house
- **'view'**- View of the house(on the scale of 0 to 4)
- **'condition'**- Condition of the house(on the scale of 0 to 5)
- **'sqft_above'**- Area above the ground
- **'sqft_basement'**- Area of the basement
- **'yr_built'**- Year in which the house was built
- **'yr_renovated'**- Year in which the house was renovated
- **'street','city','statezip','county'**- indicates the address of the house

Data Preparation:

We will transform the data. By getting rid of missing data and removing some columns. First we will create a list of column names that we want to keep or retain. Next we drop or remove all columns except for the columns that we want to retain. Finally we drop or remove the rows that have missing values from the data set.

Model Selection:

- While creating a machine learning model, we need two dataset, one for training and other for testing. But now we have only one. So let's split this in two with a ratio of 80:20. We will also divide the data frame into feature column and label column
- Here we imported `train_test_split` function of `sklearn`. Then use it to split the dataset. Also, `test_size = 0.2`, it makes the split with 80% as train dataset and 20% as test dataset.
- The `random_state` parameter seeds random number generator that helps to split the dataset.
- The function returns four datasets. Labelled them as `train_x`, `train_y`, `test_x`, `test_y`. If we see shape of this datasets we can see the split of dataset.
- We used Random Forest Regressor, which fits multiple decision tree to the data. Finally I train the model by passing `train_x`, `train_y` to the `fit` method.
- Once the model is trained, we need to Test the model. For that we will pass `test_x` to the `predict` method.



Analyze and Prediction:

In the actual dataset, we chose only 11 features:

- **'price'**- Price of the house
- **'street','city','statezip','county'**- indicates the address of the house
- **'bedrooms'**- No. of bedrooms in the house
- **'bathrooms'**-No. of bathrooms in the house
- **'sqft_living'**- Area of the plot used for living
- **'sqft_lot'**- Total area of plot
- **'floors'**- No. of floors in the house
- **'waterfront'**- No. of waterfronts in the house
- **'sqft_above'**- Area above the ground
- **'sqft_basement'**- Area of the basement

Accuracy on test set:

We got an accuracy of 88% on test set.

Saving the Trained Model:

- Once you're confident enough to take your trained and tested model into the production-ready environment, the first step is to save it into a .h5 or .pkl file using a library like pickle.
- Make sure you have pickle installed in your environment.
- Next, let's import the module and dump the model into .pkl file.
- Finally, the trained model will be deployed in a real-world scenario to detect Real Estate Price using the Web framework flask.

9. CONCLUSION

❖ Buying your own house is what every human wish for. Using this proposed model, we want people to buy houses and real estate at their rightful prices and want to ensure that they don't get tricked by sketchy agents who just are after their money. Additionally, this model will also help Big companies by giving accurate predictions for them to set the pricing and save them from a lot of hassle and save a lot of precious time and money. Correct real estate prices are the essence of the market and we want to ensure that by using this model. Experimental results show that the proposed Random Forest Regressor model gives better result than the existing system; it has the lowest error and performed very well for both training and testing data. Our proposed system achieved an accuracy of 88%. We hope this study will help to provide some methodological and practical contributions to property appraisal and presenting an alternative way to deal with the valuation of housing costs.

❖ In conclusion, the project "Real Estate Price Prediction using Random Forest Regressor" is a promising application of machine learning to the field of real estate. By using a Random Forest Regressor model, it is possible to accurately predict the prices of real estate properties based on a variety of relevant features, such as location, square footage, and number of bedrooms.

❖ One of the key advantages of using a Random Forest model is its robustness to noise and outliers, which is particularly important in the real estate industry, where data can be highly variable. Additionally, the interpretability of the model allows for greater understanding of the factors that are driving the prediction, which can be useful for decision-making.

❖ While there are potential disadvantages to using machine learning models in real estate, such as the complexity and potential for overfitting, the benefits of using a Random Forest Regressor model make it a valuable tool for predicting real estate prices. With further development and refinement, machine learning models could play an increasingly important role in the real estate industry.

10. FUTURE ENHANSEMENT

- ❖ The future direction of research may consider a bigger geological area with more features, incorporating additional property transaction data beyond housing development.
- ❖ There are several potential areas of future work for the project "Real Estate Price Prediction using Random Forest Regressor", including:
- ❖ Incorporating additional data sources: While the current project utilizes a dataset of real estate properties, there are other sources of data that could be incorporated into the model, such as economic data, crime rates, or demographic information. By incorporating these additional features, it may be possible to improve the accuracy of the predictions.
- ❖ Fine-tuning the model hyperparameters: The Random Forest Regressor model used in this project has several hyperparameters that can be adjusted to optimize its performance. Future work could involve fine-tuning these hyperparameters to achieve even better performance.
- ❖ Comparing with other models: While the Random Forest Regressor model is a powerful tool for real estate price prediction, it is possible that other machine learning models could perform even better. Future work could involve comparing the performance of the Random Forest Regressor with other models, such as neural networks or gradient boosting.
- ❖ Evaluating the impact of different features: The feature importances provided by the Random Forest Regressor model can be used to identify the most important factors that are driving the prediction. Future work could involve further evaluating the impact of these features on real estate prices, and exploring how they vary across different regions or property types.
- ❖ Incorporating real-time data: Real estate prices can change rapidly, and incorporating real-time data into the model could improve its accuracy. Future work could involve developing a system for automatically updating the model with new data as it becomes available.

11. APPENDIX

11.1 SAMPLE CODING

```
import numpy as np
import pandas as pd
from flask import Flask, request, jsonify, render_template, redirect, flash, send_file
from sklearn.preprocessing import MinMaxScaler
from werkzeug.utils import secure_filename
import pickle
app = Flask(__name__) #Initialize the flask App
price = pickle.load(open('price.pkl','rb'))
scaler = MinMaxScaler()
@app.route('/')
@app.route('/index')
def index():
    return render_template('index.html')
    #@app.route('/future')
#def future():
#return render_template('future.html')
@app.route('/login')
def login():
    return render_template('login.html')
@app.route('/upload')
def upload():
    return render_template('upload.html')
@app.route('/preview',methods=["POST"])
def preview():
    if request.method == 'POST':
        dataset = request.files['datasetfile']
        df = pd.read_csv(dataset,encoding = 'unicode_escape')
        df.set_index('Id', inplace=True)
        return render_template("preview.html",df_view = df)
#@app.route('/home')
#def home():
```

```

# return render_template('home.html')

@app.route('/prediction.html', methods = ['GET', 'POST'])
def prediction():
    return render_template('prediction.html')

<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="utf-8">

    <meta content="width=device-width, initial-scale=1.0" name="viewport">

<title></title>

    <meta content="" name="description">

    <meta content="" name="keywords">

<!-- Favicons -->

    <link href="assets/img/favicon.png" rel="icon">

    <link href="assets/img/apple-touch-icon.png" rel="apple-touch-icon">

<!-- Google Fonts -->

    <link
href="https://fonts.googleapis.com/css?family=Open+Sans:300,300i,400,400i,600,600i,700,700i|Raleway:300,300i,400,400i,500,500i,600,600i,700,700i|Poppins:300,300i,400,400i,500,500i,600,600i,700,700i" rel="stylesheet">

<!-- Vendor CSS Files -->

    <link href="../../static/vendor/aos/aos.css" rel="stylesheet">

    <link href="../../static/vendor/bootstrap/css/bootstrap.min.css" rel="stylesheet">

    <link href="../../static/vendor/bootstrap-icons/bootstrap-icons.css" rel="stylesheet">

    <link href="../../static/vendor/boxicons/css/boxicons.min.css" rel="stylesheet">

```

```

<link href="../static/vendor/glightbox/css/glightbox.min.css" rel="stylesheet">

<link href="../static/vendor/swiper/swiper-bundle.min.css" rel="stylesheet">

<!-- Template Main CSS File -->

<link href="../static/css/style.css" rel="stylesheet">

<!--

* Template Name: Kelly - v4.0.1

* Template URL: https://bootstrapmade.com/kelly-free-bootstrap-cv-resume-html-template/

* Author: BootstrapMade.com

* License: https://bootstrapmade.com/license/

-->

</head>

<body>

<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="utf-8">

  <meta content="width=device-width, initial-scale=1.0" name="viewport">

  <title> </title>

  <meta content="" name="description">

  <meta content="" name="keywords">

body {

background-image: url("../static/images/email3.jpg");

background-color: #cccccc;

}

```

```

</style>

<script>

addEventListener("load", function () {

setTimeout(hideURLbar, 0);

}, false);

function hideURLbar() {

window.scrollTo(0, 1);

}

function login(){

var uname = document.getElementById("uname").value;

var pwd = document.getElementById("pwd").value;

if(uname == "admin" && pwd == "admin")

{

alert("Login Success!");

window.location = "{{url_for('upload')}}";

return false;

}

else

{

alert("Invalid Credentials!")

}

}

<!-- Favicons -->

<link href="assets/img/favicon.png" rel="icon">

```



```

<link href="assets/img/apple-touch-icon.png" rel="apple-touch-icon">

<!-- Google Fonts -->

<!-- Vendor CSS Files -->

<link href="../static/vendor/aos/aos.css" rel="stylesheet">

<link href="../static/vendor/bootstrap/css/bootstrap.min.css" rel="stylesheet">

<link href="../static/vendor/bootstrap-icons/bootstrap-icons.css" rel="stylesheet">

<link href="../static/vendor/boxicons/css/boxicons.min.css" rel="stylesheet">

<link href="../static/vendor/glightbox/css/glightbox.min.css" rel="stylesheet">

<link href="../static/vendor/swiper/swiper-bundle.min.css" rel="stylesheet">

</head>

</body>

</html>

#loading {

    background: url('../static/ajax-loader.gif') no-repeat center center;

    position: absolute;

    top: 0;

    left: 99;

    height: 100%;

    width: 90%;

    z-index: 9999999;

!-- Vendor JS Files -->

<script src="../static/vendor/aos/aos.js"></script>

<script src="../static/vendor/bootstrap/js/bootstrap.bundle.min.js"></script>

<script src="../static/vendor/glightbox/js/glightbox.min.js"></script>

```

```

<script src="../../static/vendor/isotope-layout/isotope.pkgd.min.js"></script>

<script src="../../static/vendor/php-email-form/validate.js"></script>

<script src="../../static/vendor/purecounter/purecounter.js"></script>

<script src="../../static/vendor/swiper/swiper-bundle.min.js"></script>

<script src="../../static/vendor/waypoints/noframework.waypoints.js"></script>

<!-- Template Main JS File -->

<script src="../../static/js/main.js"></script>

</body>

</html>

@app.route('/upload')

#def upload_file():

#    return render_template('BatchPredict.html')

@app.route('/predict',methods=['POST'])

def predict():

    int_feature = [x for x in request.form.values()]

    print(int_feature)

    int_feature = [float(i) for i in int_feature]

    final_features = [np.array(int_feature)]

    prediction = price.predict(final_features)

    output = float(prediction[0])

    print(output)

    result =float(output) * float(output) * float(output)

    return render_template('prediction.html', prediction_text= int(result))

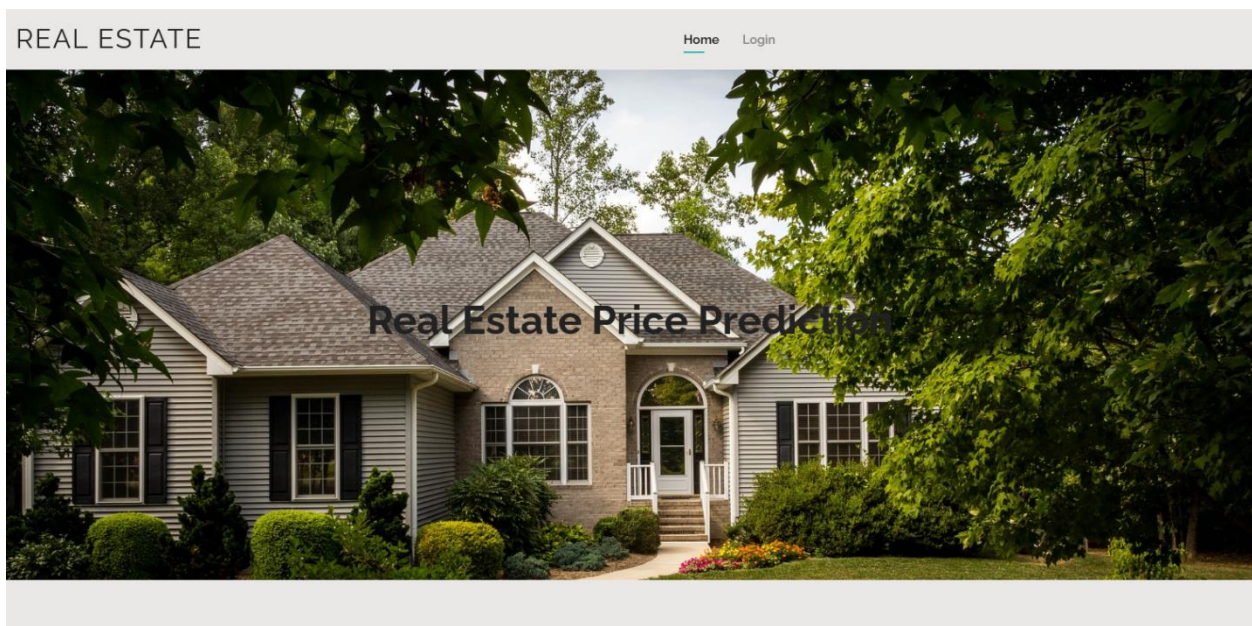
@app.route('/chart')

```

```
def chart():  
  
    return render_template('chart.html')  
  
if __name__ == "__main__":  
  
    app.run(debug=True)
```

11.2 SCREENSHOTS

HOME PAGE



LOGIN PAGE

REAL ESTATE

[Home](#) [Login](#) [upload](#) [prediction](#) [analysis](#)



Username

Username

Password

password

Login

LOGIN PAGE 2

REAL ESTATE

[Home](#) [Login](#) [upload](#) [prediction](#) [analysis](#)



Username

admin

Password

Login

PREVIEW

REAL ESTATE

[Home](#) [Login](#) [upload](#) [prediction](#) [analysis](#)

PREVIEW

	price	bedrooms	bathrooms	sqft_living	sqft_lot	floors	waterfront	view	condition	sqft_above	sqft_basement	yr_built	yr_renovated
Id													
1	3.130000e+05	3	1.50	1340	7912	1.5	0	0	3	1340	0	1955	2005
2	2.384000e+06	5	2.50	3650	9050	2.0	0	4	5	3370	280	1921	0
3	3.420000e+05	3	2.00	1930	11947	1.0	0	0	4	1930	0	1966	0

PREVIEW 2

REAL ESTATE

[Home](#) [Login](#) [upload](#) [prediction](#) [analysis](#)

ms	sqft_living	sqft_lot	floors	waterfront	view	condition	sqft_above	sqft_basement	yr_built	yr_renovated	street	city	state	zip	country
	1340	7912	1.5	0	0	3	1340	0	1955	2005	18810 Densmore Ave N	Shoreline	WA	98133	USA
	3650	9050	2.0	0	4	5	3370	280	1921	0	709 W Blaine St	Seattle	WA	98119	USA
	1930	11947	1.0	0	0	4	1930	0	1966	0	26206-26214 143rd Ave SE	Kent	WA	98042	USA
	2000	8030	1.0	0	0	4	1000	1000	1963	0	857 170th Pl NE	Bellevue	WA	98008	USA
	1940	10500	1.0	0	0	4	1140	800	1976	1992	9105 170th Ave NE	Redmond	WA	98052	USA
	880	6380	1.0	0	0	3	880	0	1938	1994	522 NE 88th St	Seattle	WA	98115	USA

TRAIN & TEST

REAL ESTATE													
HomeLoginuploadpredictionanalysis													
4598	4.169042e+05	3	2.50	3010	7014	2.0	0	0	3	3010	0	2009	0
4599	2.034000e+05	4	2.00	2090	6630	1.0	0	0	3	1070	1020	1974	0
4600	2.206000e+05	3	2.50	1490	8102	2.0	0	0	4	1490	0	1990	0

Click to Train | Test

PREDICTION

REAL ESTATE													
HomeLoginuploadpredictionAnalysis													

PREDICTION

country:USA

city:Seattle

sqft_living:2710

sqft_lot:4500

floors:2

bedrooms:3

bathrooms:2

waterfront:0

sqft_above:1880

sqft_basement:830

Predict

Prediction is:\$

PREDICTION 2

REAL ESTATE

HomeLoginuploadpredictionAnalysis

PREDICTION

country:

USA

city:

Shoreline

sqft_living:

sqft_living

sqft_lot:

sqft_lot

floors:

1

bedrooms:

0

bathrooms:

0

waterfront:

0

sqft_above:

sqft_above

sqft_basement:

sqft_basement

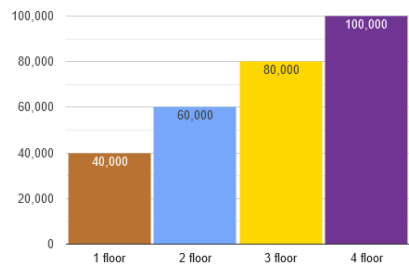
Predict

Prediction is: \$808313

CHART

CHART

Real Estate



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